

Stefan Clarke

Education

- 2021–2026 **PhD in Operations Research and Financial Engineering**, Princeton University.
Advised by Prof. Bartolomeo Stellato.
Courses: *Linear and Nonlinear Optimization, Convex Optimization, Deep Learning Theory, Theory of Reinforcement Learning, Probability, Stochastic Calculus, Statistics.*
- 2017–2021 **MMath in Mathematics and Statistics**, University of Oxford, St John's College.
First Class Honours.
Part C: 84% (*graduated first in cohort*), Part B: 82% (*fourth in cohort*).
Courses: *Statistical ML, Computational Statistics, Applied Probability, Information Theory, Martin-gales, SDEs, Large Deviations, Analysis, Complex Analysis, Linear Algebra, and many more.*
- 2010–2017 **The Nelson Thomlinson School**, Wigton, Cumbria, United Kingdom.
A-levels: *A*A*A*A*A in Mathematics, Further Mathematics, Chemistry, Biology, General Studies.*

Research Papers

- 2026 **Conformal prediction for early stopping in mixed integer optimization**, arXiv:—.—. With Bartolomeo Stellato.
Learn to speed up integer optimization solvers by early-stopping with probabilistic guarantees on optimality.
- 2025 **Learning-Based Hierarchical Approach for Fast Mixed-Integer Optimization**, arXiv:2512.03547.
With Bartolomeo Stellato.
Learn to solve parametric families of integer programs in a hierarchial fashion with probabilistic guarantees on optimality.
- 2024 **Differentiable Cutting Plane Layers for Mixed Integer Optimization**, arXiv:2311.03350.
With Gabriele Dragotto, Jaime Fernandez-Fisac, Bartolomeo Stellato.
Learn to solve parametric families of integer programs with cutting planes.
- 2024 **Maximim Shannon Capacity of Photonic Structures**, Nature Nanophotonics.
With Alessio Amaolo, Pengning Chao, Alejandro W. Rodriguez, et al.
Theory of optimal solutions to biconcave optimization in photonic devices.
- 2023 **Learning Rationality in Potential Games**, CDC 2023 Proceedings.
With Gabriele Dragotto, Jaime Fernandez-Fisac, Bartolomeo Stellato.
Optimization algorithm for learning rationality parameters in potential games.

Work History

- Jun–Aug 2025 **Tower Research Capital**, Quantitative Research Intern, New York City.
Work in investment strategy research.
- Jul–Sep 2020 **G-Research**, Quantitative Research Intern, London.
Research in investment strategies using Python, optimization, dynamic programming.
- Jul–Aug 2019 **University of Oxford**, Machine Learning Research Intern.
Supervised by Dr. Vinayak Abrol.
Porject researching variance collapse in neural processes for speech repair.
- Mar 2019 **Softwire**, Software Engineering Intern, London
Built database and management system using JavaScript, HTML, CSS, SQL.

Teaching Experience

- 2023–2026 **ORF498/499: Senior Independent Research Foundations**, Princeton University.
Provided mentoring and classes to 30 students writing senior theses over two years. Held weekly writing and research sessions.
- Spring 2023 **ORF307: Optimization**, Princeton University.
- Fall 2022 **ORF309: Probability and Stochastic Systems.**, Princeton University
- Fall 2022 **ORF523: Linear and Nonlinear Optimization**, Princeton University
Ran supplementary class for graduate students with little experience in optimization.

Presentations

- Dec 2026 **NEURIPS, Invited Talk**, San Diego, Learning Hierarchical Optimizers in Mixed Integer Programming.
- June 2025 **MIP Workshop**, Minneapolis, Learning Hierarchical Optimizers in MIP (poster).
- Oct 2024 **INFORMS**, Seattle, Reoptimization Methods in Integer Optimization.
- June 2024 **Princeton OLC**, Princeton, Differentiable cutting plane layers for MIP (poster).
- June 2024 **Oxford-Princeton Math-Finance Workshop 2024**, Princeton, Differentiable cutting plane layers for MIP.
- Dec 2023 **CDC**, Singapore, Learning Rationality in Potential Games.
- Oct 2023 **INFORMS**, Phoenix, Learning Rationality in Potential Games.

Awards and Scholarships

- 2025 **MIP Workshop Travel Grant.**
For the presentation of a research poster at MIP Workshop 2025.
- 2023 **CDC Young Researcher Travel Grant.**
For a research presentation at CDC 2023.
- 2021 **Gibb's Prize**, University of Oxford.
 For best exam performance in Mathematics and Statistics
- 2019 **Special Research Grant**, St John's College, Oxford.
- 2018–2020 **Casberd Scholarship**, St John's College, Oxford.
- 2013–2016 **Math Olympiads and Competitions:**
23rd UKMT BMO Distinction Medal. 4-time BMO qualifier. BMO team finalist (2014). UK Young Enterprise finalist.

Extracurricular Activities

- 2024– Founder, Princeton Graduate Student Musicians Society.
- 2022– Founder and guitarist, Scanderlous.
- 2017–2020 St John's College Boat Club: Vice Captain (2018), President (2019).
- 2020–2021 Events Secretary, The Invariants (Oxford Math Society).
- 2018–2020 Webmaster, Oxford University Rowing Clubs (Django).
- 2018–2020 Social Secretary, St. John's Mathematics Society.

Programming Languages

- Advanced Python.
- Intermediate C, C++.
- Basic SQL, JavaScript, HTML, CSS.

Interests

- Music *Guitarist (UK Grade 8) and pianist (UK Grade 6), Scanderlous band member, performed at Princeton events, founder of Princeton Graduate Jazz Ensemble.*
- Programming *Projects in RL (Haxball, Quoridor, Connect 4).*
- Running *Philadelphia Marathon (2022), Brooklyn Half (2024, 2026), Princeton Half (2024, 2025).*